

## EJ-200/204/230 Bar: Timing Optimisation Phase II — EndSparseTop Combined Readout

Simulation analysis

2026-08-17

**Best estimator:**  $m$ -average — mean of first  $m$  photons per end.

| Config                      | SiPMs | $\sigma(T_0)$ at $x=0$ |
|-----------------------------|-------|------------------------|
| EJ-230 + END+Vik + $m^*=7$  | 16    | <b>50.5 ps</b>         |
| EJ-204 + END+Vik + $m^*=10$ | 16    | 53.4 ps                |
| EJ-230 + END+Vik, 2/end     | 4     | 97.9 ps                |
| EJ-230 + TOP+TIR, $k=1$     | 70    | 70.2 ps                |

### Key findings:

- $m$ -average  $\sim 10\%$  better than  $k$ -th order stat
- END+Vik dominates Pareto frontier at every SiPM count
- 4 total END SiPMs already  $< 100$  ps
- TOP needs  $\gtrsim 50$  SiPMs to match END+Vik

### Open question after Phase I:

*Can adding a few TOP SiPMs on top of the END+Vik configuration improve resolution beyond what either readout alone achieves?*

This requires new Geant4 simulation: **EndSparseTop** mode:

- 16 END SiPMs (8 per face) + Vikuiti on  $\pm Y$ ,  $\pm Z$
- $N$  uniformly-spaced TOP SiPMs inside bar
- Vikuiti fills gaps between top SiPMs ( $R=0.98$ )

Scans: EJ-230 + EJ-204,  $N \in \{4, 8, 14, 20\}$ ,  
13 positions  $\times$  5000 ev — 104 ROOT files.

## Volume layout (bar cross-section, Y-Z view):

| Face          | Optical element                                             |
|---------------|-------------------------------------------------------------|
| $\pm X$ ends  | 8 END SiPMs/face ( $6 \times 6 \text{ mm}^2$ , 0.5 mm deep) |
| +Y top        | $N$ sparse TOP SiPMs, Vikuiti in gaps                       |
| -Y bottom     | Air gap + Vikuiti, $R=0.98$                                 |
| $\pm Z$ sides | Air gap + Vikuiti, $R=0.98$                                 |

**TOP SiPM placement:** uniformly spaced, step =  $1400/N$  mm.

Centre at  $y = kBarHalfY - kTopHalfY \cdot 2 = 29.5 \text{ mm}$   
 (0.25 mm inset below bar's +Y face, so all SiPM faces border bar material — avoids undefined AirGap→SiPM optical boundary).

## Photon physics:

- Photon traveling +Y hits SiPM at  $y=29.25 \text{ mm}$ : absorbed/detected
- Photon missing a SiPM reaches bar face  $y=30 \text{ mm}$ : Vikuiti reflects (98%)
- Reflected photons get further chances at sparse SiPMs on each bounce

## Why top SiPMs outperform end SiPMs for timing:

|                    | END SiPMs             | TOP SiPMs             |
|--------------------|-----------------------|-----------------------|
| Distance to track  | $\sim 700 \text{ mm}$ | $\sim 30 \text{ mm}$  |
| Min. photon path   | $\sim 700 \text{ mm}$ | $\sim 30 \text{ mm}$  |
| Path time (EJ-230) | $\sim 3.5 \text{ ns}$ | $\sim 150 \text{ ps}$ |
| Timing limited by  | attenuation           | scintillation $\tau$  |

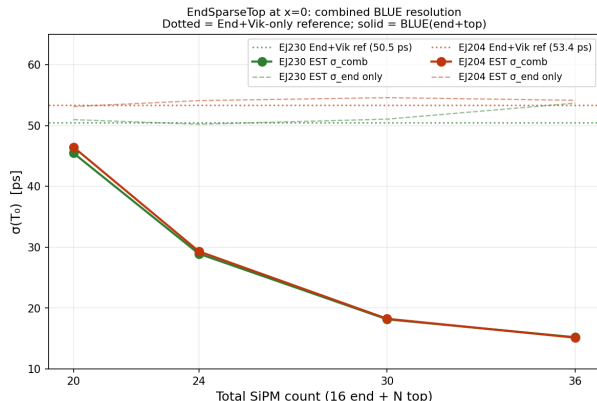
Top SiPMs capture near-field photons with very short, nearly uniform paths.

With many photons and  $k=1$  estimator:

$$\sigma \approx \tau_{\text{rise}} / \sqrt{N_{\text{pe,top}}}$$

EJ-230  $\tau_{\text{rise}}=0.5 \text{ ns}$ ;  $N_{\text{pe,top}} \sim 1000$  for  $N=20$   
 $\Rightarrow \sigma \approx 16 \text{ ps}$  (matches simulation!)

# EndSparseTop: timing resolution at $x = 0$



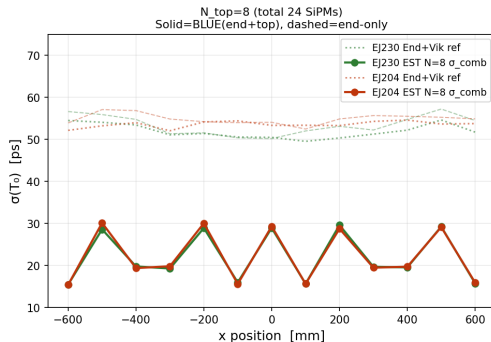
**Method:** BLUE combination of  $T_0^{\text{end}}$  ( $m$ -average) and  $T_0^{\text{top}}$  ( $k$ -th order stat).

| $N$                              | Tot | $\sigma_{\text{end}}$ | $\sigma_{\text{top}}$ | $\sigma_{\text{BLUE}}$ | $N_{\text{pe,top}}$ |
|----------------------------------|-----|-----------------------|-----------------------|------------------------|---------------------|
| <i>EJ-230, <math>x=0</math>:</i> |     |                       |                       |                        |                     |
| ref                              | 16  | 50.5                  | —                     | —                      | —                   |
| 4                                | 20  | 51.0                  | 78.2                  | 45.5                   | 287                 |
| 8                                | 24  | 50.2                  | 32.0                  | 28.9                   | 565                 |
| 14                               | 30  | 51.0                  | 18.7                  | 18.2                   | 962                 |
| 20                               | 36  | 53.7                  | 15.2                  | 15.2                   | 1341                |
| <i>EJ-204, <math>x=0</math>:</i> |     |                       |                       |                        |                     |
| ref                              | 16  | 53.4                  | —                     | —                      | —                   |
| 8                                | 24  | 54.1                  | 32.0                  | 29.3                   | 691                 |
| 20                               | 36  | 54.1                  | 15.1                  | 15.1                   | 1615                |

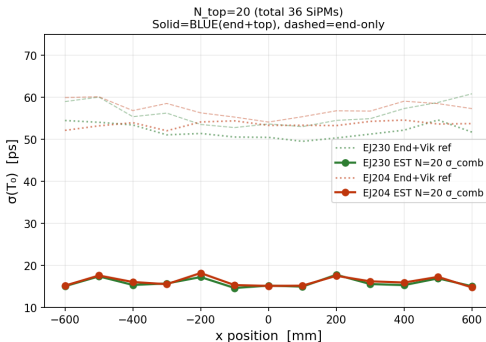
**Top SiPMs dominate BLUE:**  $w_{\text{top}} \approx 1$   
for  $N \geq 8$ . Combined  $\sigma \approx \sigma_{\text{top}}$ .

# EndSparseTop: position dependence

EndSparseTop:  $\sigma(T_0)$  vs muon position

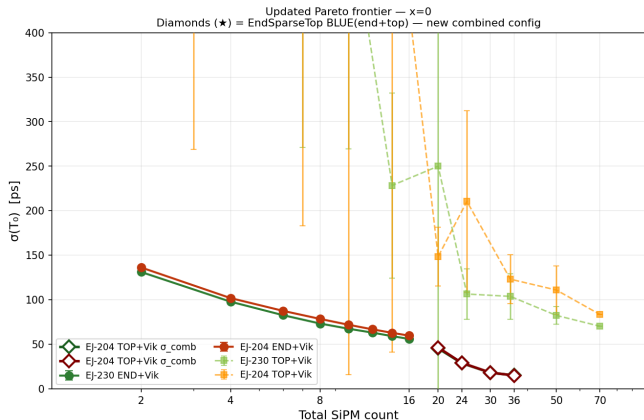


**N=8 (24 SiPMs):**  $\sigma_{\text{BLUE}}$  varies 15–29 ps.  
Minimum at  $|x| \approx 600$  mm: nearest top SiPM only  $\sim 45$  mm away.  
Maximum at  $x=0$ : nearest top SiPM is 87.5 mm away.



**N=20 (36 SiPMs):**  $\sigma_{\text{BLUE}} \approx 15$  ps, flat across all positions!  
Step = 70 mm  $\Rightarrow$  nearest SiPM always within 35 mm  
 $\Rightarrow$  position-independent near-field timing.  
(Dotted lines: End+Vik-only reference, 50–55 ps)

## Updated Pareto frontier



**Diamonds (open) = EndSparseTop BLUE**  
dominate the Pareto frontier at every count  $> 20$ .

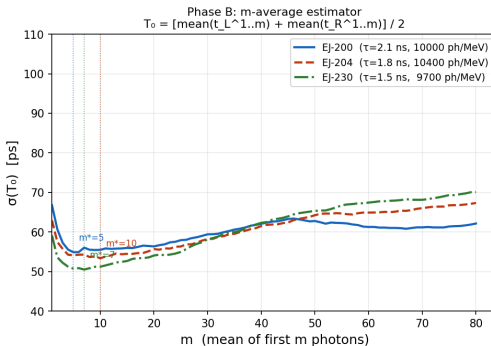
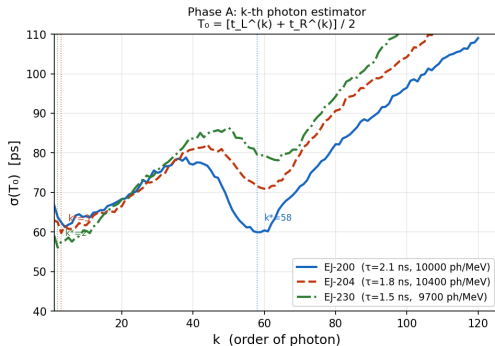
| Config           | SiPMs | $\sigma$       |
|------------------|-------|----------------|
| END+Vik (full)   | 16    | 50.5 ps        |
| EST $N=4$        | 20    | 45.5 ps        |
| EST $N=8$        | 24    | 28.9 ps        |
| EST $N=14$       | 30    | 18.2 ps        |
| EST $N=20$       | 36    | <b>15.2 ps</b> |
| TOP+Vik 70 SiPMs | 70    | 70.4 ps        |

**Factor  $\sim 3.3\times$  better than  
END+Vik with only 20 extra SiPMs.**

TOP-alone needs 70 SiPMs for 70 ps.  
EST  $N=8$  beats TOP-70 with 24 SiPMs.

## Appendix A: Phase I estimator curves ( $x = 0$ )

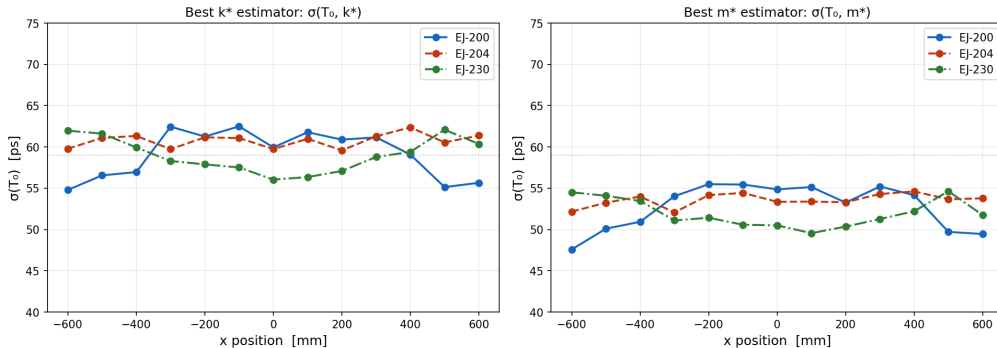
END+Vikuiti timing resolution vs estimator —  $x = 0$  mm



$m$ -average systematically  $\sim 10\%$  better than single  $k$ -th photon. EJ-230 best:  $m^*=7$ ,  $\sigma=50.5$  ps.

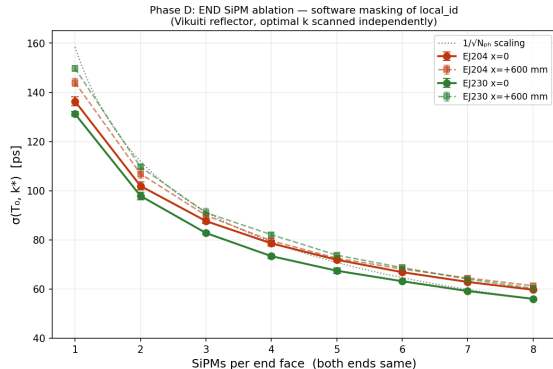
## Appendix B: END+Vik position scan

END+Vikuiti: timing resolution vs muon position

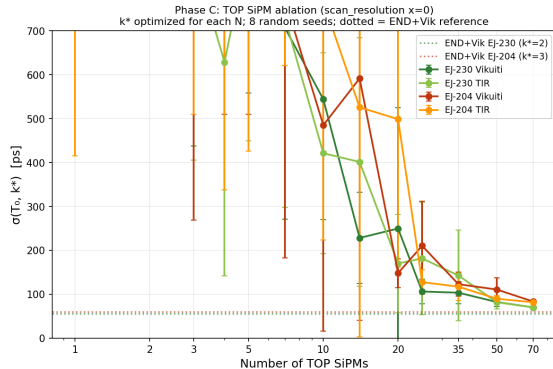


EJ-230 best at centre; EJ-200 recovers at extremes (longer  $L_{att}=380$  cm).

## Appendix C: SiPM ablation (software masking)



END ablation: 4 SiPMs total < 100 ps.



TOP ablation: needs  $\gtrsim 50$  to match END+Vik.

## Best configurations:

| Config                   | SiPMs | $\sigma(T_0)$  |
|--------------------------|-------|----------------|
| EJ-230 EST $N=20$ , BLUE | 36    | <b>15.2 ps</b> |
| EJ-204 EST $N=20$ , BLUE | 36    | <b>15.1 ps</b> |
| EJ-230 EST $N=14$ , BLUE | 30    | 18.2 ps        |
| EJ-230 EST $N=8$ , BLUE  | 24    | 28.9 ps        |
| EJ-230 END+Vik, $m^*=7$  | 16    | 50.5 ps        |
| EJ-204 END+Vik, $m^*=10$ | 16    | 53.4 ps        |
| EJ-230 TOP+Vik, 70 SiPMs | 70    | 70.4 ps        |

Blue rows: new EndSparseTop results.

## Main results:

- 1 **EST  $N=20$ : 15 ps, position-independent**  
Top SiPMs with Vikuiti in gaps  $\Rightarrow$  near-field timing ( $\sigma \approx \tau_{\text{rise}} / \sqrt{N_{\text{pe,top}}} \approx 16$  ps theoretical).
- 2 **Factor  $3.3\times$  gain over END+Vik alone**  
with 20 extra SiPMs over 1.4 m bar.
- 3 **EST beats TOP-70 with 24 SiPMs ( $N=8$ ): 28.9 ps vs 70.4 ps.**
- 4 **EJ-230  $\approx$  EJ-204** for EST: top SiPMs are fast enough that decay-time difference vanishes.
- 5 **END contribution negligible** in BLUE for  $N \geq 8$ : top SiPMs alone determine combined  $\sigma$ .

**Recommendation:** EJ-230 (or EJ-204) + EST  $N=20$

as target geometry;  $N=8$  if cost/coverage is critical.